

The University of Texas at Austin
IN-CLASS WORK 5

M339G Predictive Analytics

October 05, 2026

LOGISTIC REGRESSION.

Problem 5.1. You run a logistic regression to predict the binary response Y , indicating whether a credit card holder will default on a payment in the next six months, from a single numerical explanatory variable X (number of late payments in the past year).

You are given the following estimated coefficients:

- $\hat{\beta}_0 = -3.5$
- $\hat{\beta}_1 = 0.8$

Calculate the fitted **odds** for a card holder with three late payments. Then, convert the odds to the corresponding **probability of success**.

Solution. In the logistic model, the odds are $e^{\beta_0 + \beta_1 x}$. So, the coefficients given above yield

$$\text{fitted odds} = e^{-3.5 + 0.8(3)} = e^{-1.1}.$$

Thus, the fitted odds of success at $X = 3$ are approximately

$$e^{-1.1} \approx 0.33.$$

To convert the odds to a probability, we use

$$p(x) = \frac{\text{odds}(x)}{1 + \text{odds}(x)}.$$

Hence,

$$p(3) = \frac{0.33}{1 + 0.33} \approx 0.25.$$

In words, this means that a card holder with three late payments has about a 25% chance of defaulting in the next six months according to the fitted model.

Problem 5.2. An insurer uses logistic regression to model the probability that a policyholder files a claim in the next year, using a single numerical explanatory variable X (age of the insured vehicle, in years). The estimated slope coefficient is positive.

Determine whether each of the following statements is **TRUE** or **FALSE**. Justify each answer in one or two sentences.

1. The fitted probability of filing a claim increases as the age of the vehicle increases.
2. Each additional year of vehicle age increases the fitted probability of filing a claim by the same amount.
3. Each additional year of vehicle age multiplies the fitted odds of filing a claim by the same factor.
4. The fitted probabilities always lie strictly between 0 and 1, regardless of the value of X .

5. Ordinary least squares regression of Y on X would be an equally appropriate model for these data.

Solution.

1. **TRUE.** The fitted probability $p(x) = \frac{e^{\beta_0 + \beta_1 x}}{1 + e^{\beta_0 + \beta_1 x}}$ is an increasing function of x whenever $\beta_1 > 0$.
2. **FALSE.** The logistic curve is S-shaped, so the change in probability per unit change in x depends on the current value of x . It is largest near $p(x) = 0.5$ and smallest in the tails.
3. **TRUE.** The odds are $e^{\beta_0 + \beta_1 x}$, so increasing x by one unit multiplies the odds by e^{β_1} , regardless of the starting value of x .
4. **TRUE.** Since $e^{\beta_0 + \beta_1 x} > 0$, the fitted probability is of the form $\frac{a}{1+a}$ with $a > 0$, which always lies strictly between 0 and 1.
5. **FALSE.** The response is binary. A linear fit can produce predicted values below 0 or above 1, which cannot be interpreted as probabilities, and it assumes a constant change in probability per unit of x .

Problem 5.3. You are using logistic regression to predict the probability that a life insurance policy will lapse in the next year. Your predictor is a categorical random variable indicating the *Payment* frequency: *Monthly*, *Quarterly*, *Annual*. The *Monthly* category is understood as the *baseline*. You obtain the following summary of coefficients:

Intercept	-1.842
PaymentQuarterly	-0.613
PaymentAnnual	-1.307

What is the fitted probability that a policy with *Quarterly* payments will lapse?

Solution.

$$\hat{p} = \frac{e^{-1.842 - 0.613}}{1 + e^{-1.842 - 0.613}} = 0.07907368.$$

Problem 5.4. You are using logistic regression to predict the probability of a particular class of driver having a claim in the next insurance period. Your predictors are two categorical random variables indicating the

- *Region* in which the driver lives: *North*, *South*, *East*, *West* with *North* as *baseline*.
- *Use of the vehicle*: *Personal*, *Commute*, *Business* with *Personal* as the *baseline*.

You obtain the following summary of coefficients:

Intercept	-1.732
RegionSouth	0.128
RegionEast	-0.064
RegionWest	0.211
UseCommute	0.342
UseBusiness	0.587

What is the fitted probability that a driver from the *West* region who uses the vehicle for *Business* will have a claim?

Solution.

$$\hat{p} = \frac{e^{-1.732+0.211+0.587}}{e^{-1.732+0.211+0.587} + 1} = 0.2821139$$

Problem 5.5. In a study, 200 customers were asked to choose one of three insurance plans (Basic, Standard, Premium). The customers were organized into four income categories (Low, Lower-middle, Upper-middle, High).

A logistic regression was fitted to the customers' responses to predict their preferred plan with income category Low and the Basic plan as reference categories.

For income category Low the log-odds for preference of the Standard plan and the Premium plan were -0.412 and -1.127 , respectively.

Calculate the modeled probability of someone from income category Low preferring the Premium plan.

Solution. Since the log-odds are given to be -1.127 , the odds are $e^{-1.127}$. Hence, the modeled probability is

$$\hat{p} = \frac{e^{-1.127}}{1 + e^{-0.412} + e^{-1.127}} = 0.1631170.$$

Problem 5.6. A statistician uses a logistic model to predict the probability of success, π , of a binomial random variable.

You are given the following information:

- There is one predictor random variable, X , and an intercept in the model.
- The estimates of π at $x = 2$ and $x = 5$ are 0.57444 and 0.92757 , respectively.

Calculate the estimated intercept coefficient, b_0 , and the slope coefficient, b_1 , in the logistic model that produced the above probability estimates.

Solution. To obtain the fitted regression, we apply the *logit* function to the given probabilities.

$$\begin{aligned}\ln\left(\frac{0.57444}{1-0.57444}\right) &= b_0 + b_1(2) \\ \ln\left(\frac{0.92757}{1-0.92757}\right) &= b_0 + b_1(5)\end{aligned}$$

We obtain the following system of two equations with two unknowns:

$$\begin{aligned}b_0 + 2b_1 &= 0.299990 \\ b_0 + 5b_1 &= 2.549948\end{aligned}$$

Subtracting the first from the second equation above, we get

$$3b_1 = 2.549948 - 0.299990 = 2.249958 \Rightarrow b_1 = 0.749986.$$

Reusing the first equation, we obtain

$$b_0 = 0.299990 - 2(0.749986) = -1.199982.$$